

Quiz (Carolina Reaper)

- Q1.** If the slope of a line is m , what is the slope of a line perpendicular to it?
- (A) m
(B) $-m$
(C) $\frac{1}{m}$
(D) $-\frac{1}{m}$
(E) 0
- Q2.** Which of the following lines is perpendicular to $y = 2x + 1$?
- (A) $y = 2x - 1$
(B) $y = -\frac{1}{2}x + 1$
(C) $y = \frac{1}{2}x - 1$
(D) $y = -2x + 1$
(E) $y = x + 1$
- Q3.** What is the equation of a line perpendicular to $y = 3x - 2$ and passing through $(1, 4)$?
- (A) $y = -\frac{1}{3}x + 4$
(B) $y = -\frac{1}{3}x + \frac{7}{3}$
(C) $y = \frac{1}{3}x + 4$
(D) $y = 3x - 2$
(E) $y = -3x + 4$
- Q4.** If two lines have slopes m_1 and m_2 , what condition must be met for them to be perpendicular?
- (A) $m_1 = m_2$
(B) $m_1 = -m_2$
(C) $m_1 \cdot m_2 = 1$
(D) $m_1 \cdot m_2 = -1$
(E) $m_1 + m_2 = 0$
- Q5.** Which of the following is an example of two perpendicular lines?
- (A) $y = 2x + 1$ and $y = 2x - 1$
(B) $y = x + 1$ and $y = -x + 1$
(C) $y = 3x + 2$ and $y = 3x - 2$
(D) $y = x - 2$ and $y = -x - 2$
(E) $y = 2x + 1$ and $y = 2x + 2$
- Q6.** What is the slope of a line perpendicular to $y = \frac{2}{3}x - 1$?
- (A) $\frac{3}{2}$
(B) $-\frac{3}{2}$
(C) $\frac{2}{3}$
(D) $-\frac{2}{3}$
(E) 0
- Q7.** If a line has a slope of -2 , what is the slope of a line perpendicular to it?
- (A) 2
(B) -2
(C) $\frac{1}{2}$
(D) $-\frac{1}{2}$
(E) 0
- Q8.** Which of the following lines is perpendicular to $y = -x + 2$?
- (A) $y = x + 2$
(B) $y = -x - 2$
(C) $y = x - 2$
(D) $y = -x + 1$
(E) $y = x + 1$

Open Ended Questions (FRQ)

- Q9.** Find the equation of a line perpendicular to $y = 2x - 3$ and passing through $(2, 5)$. Show your work and explain your reasoning.
- Q10.** Prove that the lines $y = 3x + 2$ and $y = -\frac{1}{3}x - 1$ are perpendicular. What is the point of intersection of these two lines?

Chef's Tips

- Tip 1: Make sure students understand the concept of negative reciprocal slopes.
- Tip 2: Encourage students to use graphs to visualize perpendicular lines.
- Tip 3: Emphasize the importance of showing work and explaining reasoning in free response questions.